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How does a driver perceive risk when making decision of lane-changing?

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Abstract: This paper shows a case study to show how a driver perceives driving risk in order to model decision making of changing lanes. Lane-changing data are collected under 28 driving conditions through a driving simulator. Analyses are focused on driver's risk perception to a leading vehicle driving on the host lane and nearest rear and front vehicles driving on a targeted lane. The analyses are operation on two stages when a driver decides to change lanes (i.e., turning on turn signal), and then when he or she is cutting in the target lane in order to comprehend lane-changing decision making. The results suggest that a driver tends to prefer the time to collision (TTC) to the time headway (THW) to the rear vehicle. When the host vehicle is cutting into the target lane, a driver tends to perceive the traffic condition through both TTC and THW. Meanwhile, it is also implied that a driver tries to grasp status of each of related vehicles driving around the host vehicle for executing lane-changing behavior. These findings illustrates driver's risk perception of making a decision and operating lane-changing behavior, which suggests it is valid to model lane-changing decision making via perceiving driving risk.

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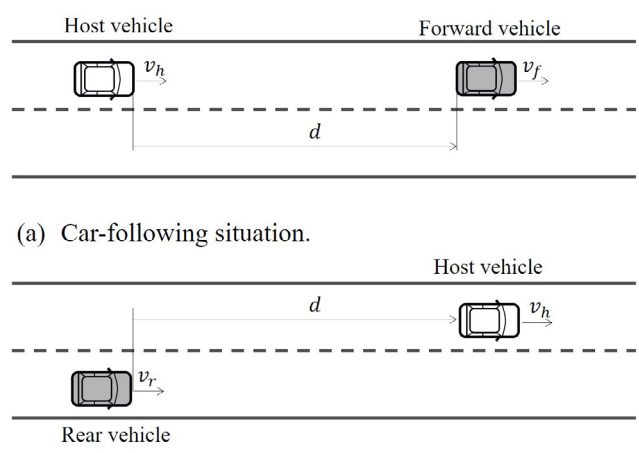
Keywords: Risk perception, Lane-changing behavior, decision making.

1. INTRODUCTION

Subjective estimate of driving risk in dynamical situation is preferred to objective facts associated with a hazard (See Rosenstock (1974), Janz and Becker (1984)). Note that driver's subjective estimate principally depends on visual information, which counts for approximately 90% of all information involving driving (See Sivak (1996)). What extent a driver perceives driving risk is greatly dependent upon what extent of information he or she obtains from a changing traffic environment, and how. The reason is that a difference exists between the magnitude of a physical stimulus and the perceived magnitude of the stimulus in a psychophysical theory (See Stevens (1961), Allan (1979)). In line with the common ground in psychophysical laws, larger the magnitude of a stimulus becomes, lower it becomes to precisely perceive the difference. Many previous studies (See Lee (1974), Lee (1976), Groeger (2000), Groeger (2002) and Kondoh et al. (2008)) have made efforts to illustrate relations between physical visual stimulus and driver's risk estimation in dynamical traffic conditions (e.g., headway distance, time headway and time-to-collision). Those studies have aimed to find a proximate measure for accurately describing the relations under typical car-following situations as shown in Fig. 1(a).

With respect to one of microscopic flow characteristics, May (1990)),Winsum and Heino (1996), Lewis-Evans et al. (2010), Risto and Martens (2013) have discussed time-headway (THW). It is originally defined as the time difference between any two consecutive vehicles when they cross a reference point (Winsum and Heino (1996)). Practically, time-headway is calculated as equation (1) in car-following situations (Fig. 1(a)). Cutting and Vishton (1995) argue the accuracy of perceiving distance through several dimensions. Their argument suggests that it is more accurate to estimate a time headway with comparison with perceiving a distance because the perceived distance is sensitive to variations in speed. Furthermore,Winsum and Heino



(a) Car-following situation.

(b) Lane-changing situation.

Fig. 1. Car-following situations and lane-changing situations

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(1996), Winsum and Brouwer (1997), Risto and Martens (2013) evaluate that drivers preferred to choice of time headway in car following and braking response.

$$THW = d/v_h \tag{1}$$

In studies of visual control of locomotion, a concept of time-to-collision (TTC) is argued as an optical flow (See Lee (1974), Lee (1976)). Hayward (1972) gives a definition of time-to-collision as: "The time required for two vehicle to collide if they continue at their present speed and on the same path". Since the visual information of time-to-collision has been proved to be a crucial predictor for estimating collision risk in car-following situations (See Cavallo and Laurent (1988), Groeger (2000), Groeger (2002)) time-to-collision is discussed in collision avoidance systems (See van der Horst (1991),van der Horst and Hogema (1993), Winsum and Heino (1996)). Generally, it is manipulated as equation (2) in cases of car-following situations.

$$TTC = d/(v_h - v_f) \tag{2}$$

In recent studies, it is pointed that driver's subjective estimation of driving risk strongly depends on the global optical flow (i.e., inverse of time-headway) and the local optical flow (i.e., inverse of time-to-collision) in cases of a host vehicle approaching to a slower forward one (See Kondoh et al. (2008), Kondoh et al. (2014)). The studies quantify the subjective estimation as *Risk Perception* (RP) according to the following formula (equation (3)).

$$RP = 1/THW + 4/TTC \tag{3}$$

Those studies of visual information in risk perception have focused on traffic accidents in the car-following situations. Accurately obtaining visual information of a rear car in lane-changing situations (e.g., Fig. 1(b)) is more difficult than a forward car in car-following situations (e.g., Fig. 1(a)). Concretely, perceiving risk is through direct and continuous visual information in cases of the forward vehicle. In cases of the rear vehicles, the visual information is indirect and discontinuous. Understanding driver's subjective perception of risk to a rear car is an essential issue to driving safety in lane-changing behavior. Consequently, this study directs driver's risk estimation to a rear vehicle that approaches to the host vehicle on the passing lane in the lane-changing situations.

The purpose of the present paper is to clarify relations of driver's estimation of collision risk to an approaching rear car in lane-changing situations to choices of time-headway, time-to-collision, RP as well as headway distance. The relations are argued in lane-changing for *choice* and lane-changing *implementation* of decision-making behaviors. From the viewpoint of a host vehicle, these measures are redefined as equation (4) - (6).

$$THW_r = -d/(-v_h) = d/v_h. \tag{4}$$

$$TTC_r = -d/(-v_h - (-v_r)) = d/(v_r - v_h). \tag{5}$$

$$RP_r = 1/THW_r + 4/TTC_r. \tag{6}$$

2. METHOD

Driver's lane-changing data are collected in different lane-changing situations operated on a driving simulator.

2.1 Apparatus

A fixed-base driving simulator is used in the data collection. It can simulate driving on an expressway with two lanes (See Fig. 2). A speed governor is set up for preventing the host vehicle from a limited velocity $v_{Max.}$.

Fig. 2. A fixed-base driving simulator.

2.2 Participants

Seven females and five males, ranging from 28 to 45 years old (Mean = 38.3, SD = 5.3), participate in the data collection. Each participant holds a valid driver's license. After receiving an explanation of the data collection, all of them signed in information consent for participating the data collection.

2.3 Driving scenario and driving task

One lane-changing scenario is designed in each trial. There are 28 trials in the whole data collection. Driving initial conditions (i.e., RP_0, v_h, v_r and d_0) in the 28 trials are sorted in terms of the index of RP as shown in Table (1).

Following instructions are given to each of participants:

(1) Each participant is requested to accelerate the host vehicle's velocity (v_h) and to reach the limited velocity $v_{Max.}$ on the cruising lane after starting a trial, where the headway distance (D) between two vehicles driving on the passing lanes is set a same value (See Fig. 3(a));
(2) he or she is given a piece of acoustic message at the time point t_0 when the third vehicle on the passing lane is passing the host vehicle (See Fig. 3(b));
(3) he or she has to make a choice whether he or she can change lanes after receiving the acoustic message. The time point is defined as t_d when he or she turns on the right turn signal for claiming he or she makes a choice to change lanes;

(4) he or she is instructed to implement the lane-change behavior as soon as possible after making the choice of changing lanes. The time point is defined as t_i when the host vehicle's body firstly touches the center line. Note that after the host vehicle passed the center line, the limited velocity of v_h is canceled.
(5) In cases of giving up a lane-changing choice, he or she is requested to turn on the left turn signal.

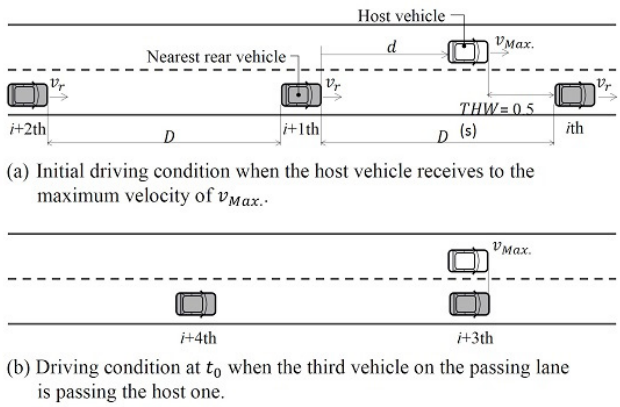

Fig. 3. Driving scenario.

2.4 Procedure

Each participant is given an explanation of the driving simulator's operation and the instructions during driving task. Then, he or she does approximately 5-minute driving exercise for being familiar with the driving simulator. After 5-minute rest, he or she is instructed to experience the 14 trials. After 5-minute rest, he or she experiences the other 14 trials. All participants are divided into two groups in order to counterbalance order effects. One group is operated in an ascending order of RP, the other is in a descending order.

After each trial, they are asked to report their subjective assessment to the experienced scenario by using a paper-questionnaire as shown in Fig. 4.

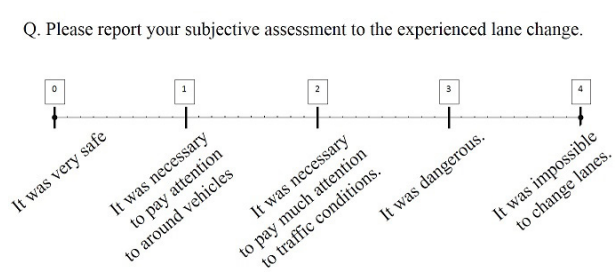

Fig. 4. Questionnaire sheet that is used for reporting subjective assessment to each scenario after each trial.

2.5 Measures

Since all participants are instructed to turn on the turn-signal once they make a choice of changing lanes in the near future, it can be considered that lane-change *choice* is made at t_d. Furthermore, t_i denotes the time point when lane-change *implementation* is initiated, i.e., the host vehicle's body firstly touches the center line. In order to investigate driver's perceiving driving risk during his/her decision making for changing lanes, this study investigates following measures at lane-change *choice* and *implementation*:

- Necessary time to decide lane changes that is time length from t_0 to t_d when the right turn signal is turned on.
- time-headway, time-to-collision, RP, and headway-distance to the nearest rear vehicle on the passing lane at t_i, which are defined as $THW_{init.}$, $TTC_{init.}$, $RP_{init.}$ and $d_{init.}$, respectively.
- Subjective assessment to each lane-change scenario

Table 1. Initial driving condition in each trial.

$No.$	RP_0	v_h	v_r	d_0	$No.$	RP_0	v_h	v_r	d_0
1	0.75	60	85	59	15	1.50	90	100	24
2	0.75	60	95	74	16	1.50	90	110	35
3	0.75	90	100	48	17	2.00	60	85	22
4	0.75	90	110	70	18	2.00	60	95	28
5	1.00	60	85	44	19	2.00	90	100	18
6	1.00	60	95	55	20	2.00	90	115	26
7	1.00	90	100	36	21	2.50	60	85	18
8	1.00	90	110	53	22	2.50	60	95	22
9	1.25	60	85	36	23	2.50	90	100	14
10	1.25	60	95	44	24	2.50	90	115	21
11	1.25	90	100	29	25	3.00	60	85	15
12	1.25	90	110	42	26	3.00	60	85	19
13	1.50	60	85	30	27	3.00	90	100	12
14	1.50	60	95	37	28	3.00	90	115	18

Unit of velocity is km/h.
Unit of distance is m.

3. RESULTS

3.1 Driver's lane-changing behavior

Fig. 5 indicates averaged horizontal position at each 1-second time period as a function of time for 5-second time period prior and 3-second time period posterior to the time point t_d. Results show that lane-changing behaviors are operated in a good manner while time-to-collision is large but time-headway is small in such trials shown in Fig. 5(a). As time-to-collision becomes smaller and time-headway becomes larger, the driving behaviors posterior to t_d tend to become in a disorder in such trials shown in Fig. 5(b) and (d). The disorder tendency becomes much clearer when both of time-to-collision and time-headway become smaller in such trials shown in Fig. 5(c). Note that the disorders also occur prior to t_d. Furthermore, host vehicle's transition in the horizontal direction while changing lanes retards as RP becomes great.

3.2 Driver's lane-changing choice

Fig. 6 plots necessary time of making a choice whether to change lanes, or not, related to each of initial driving conditions (i.e., THW_0, TTC_0, RP_0 and d_0). It is indicated that more dangerous the traffic condition is, more time is spent on make the choice. Linear regression analysis is performed between the necessary time and each of measures.

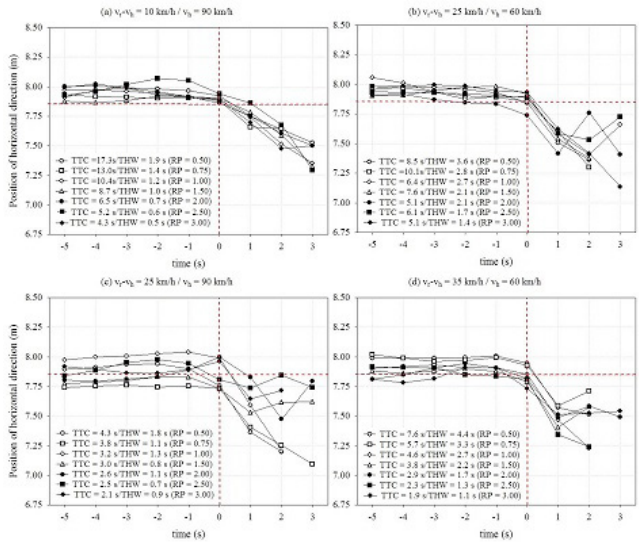

Fig. 5. Horizontal moving position of the host vehicle. Note that zero of the horizontal axis is the time point t_d at which a driver turns on the right turning signal. Each of coordinate values of Y- axis is an average horizontal position during every 1-second time period from 5 s prior and 3 s posterior to t_d for 12 participants. Dotted horizontal line denotes a center-position of the cruising lane.

Results show strongly positive correlation between TTC_0 and the necessary time ($r = 0.60$, $p < 0.01$) than THW_0 and d_0 (See Table (2)).

It is indicated that a driver tends to perceive the driving risk through time-to-collision information to an approaching rear car for making a choice whether it is executable to change lanes under the current traffic condition, or not. It is implied that driver's subjective risk estimation depends greatly on a local optical flow.

Table 2. Statistical parameters resulting from the linear regression analysis between the necessary time to make a choice and the four measures (THW_0, TTC_0, RP_0 and d_0)

	Parameters					
Measures	r^2	r	Df	MS	F	p
THW_0	0.090	0.30	26	1.560	2.569	0.12
TTC_0	0.294	0.54	26	1.21	10.81	0.002
RP_0	0.289	0.54	26	0.443	10.56	0.003
d_0	0.200	0.45	26	241.0	6.49	0.02

3.3 Driver's lane-changing implementation

Fig. 7 indicates relationships of four measures under conditions of lane-change *implementation* versus initial traffic conditions, i.e., THW_0, TTC_0, RP_0 and d_0. Linear regression analysis is operated on each of four measures between lane-change *implementation* versus traffic conditions. According to the results, strong correlations (r = 0.62 - 0.97) are observed on each of four measures (See Table (3)). Especially, greatly significant positive correlation is showed on both THW_0 ($r = 0.97$, $p < 0.01$) and TTC_0 ($r = 0.97$, $p < 0.01$).

It is argued that a driver implements lane-changing behavior while perceiving driving risk through not only time-to-collision but also time-headway to an approaching rear car. In comparison that a driver refers to perceiving only time-to-collision for lane-changing *choice*, he or she prefers to both of time-to-collision and time-headway for lane-changing *implementation*. It is suggested that a driver intends to perceive a rear-collision risk dependently on both a global optical flow and a local optical flow while he or she implements a lane change.

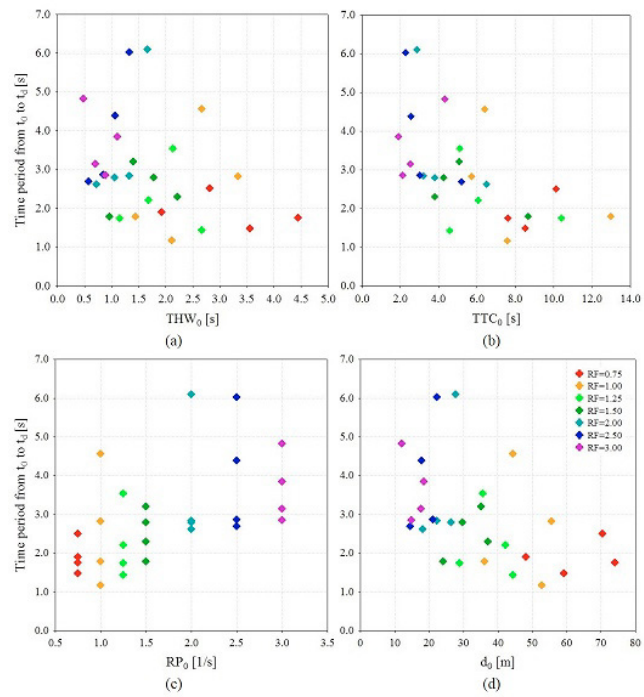

Fig. 6. Necessary time for making a choice of lane-changing behavior

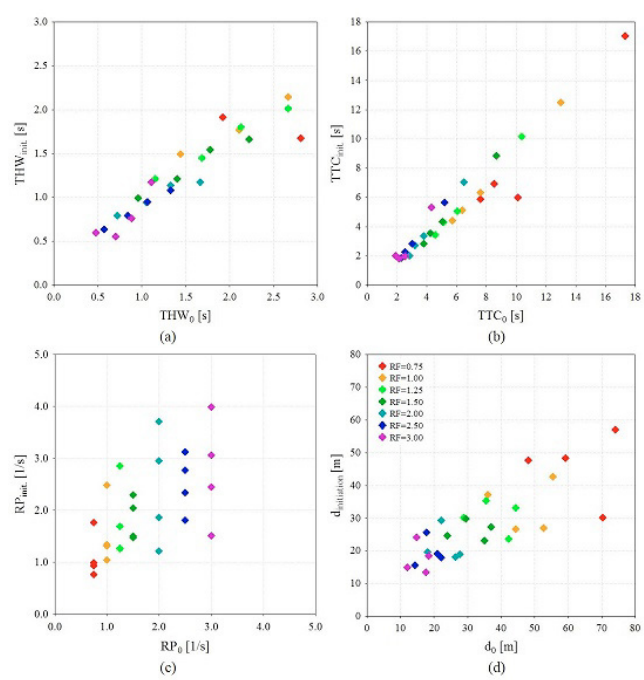

Fig. 7. Relationship of four measures between lane-change implementation and initial traffic conditions.

Table 3. Statistical parameters resulting from the linear regression analysis on four measures (time-headway, time-to-collision, RP and headway-distance) between lane-change *implementation* and initial traffic conditions

	Parameters					
Measures	r^2	r	Df	MS	F	p
THW_0	0.95	0.97	26	0.052	475.3	0.00
TTC_0	0.93	0.96	26	0.894	363.8	0.00
RP_0	0.385	0.62	26	0.384	16.26	0.0004
d_0	0.607	0.78	26	118.5	40.09	0.00

3.4 Subjective assessment

Mean subjective assessment to each trial is plot to each of four measures of initial traffic conditions, i.e., THW_0, TTC_0, RP_0 and d_0 in Fig. 8. Higher the score is, more dangerous the scenario is thought. Results in Fig. 8 show a same tendency on all the four measures. More definite correlation is indicated between the subjective assessment and values of TTC_0. It is argued that a driver prefers visual information of time-to-collision for evaluating what extent the situation is, that is, visual stimulus in the local optical flow field affects greatly on his/her subjective assessment to the rear-collision risk.

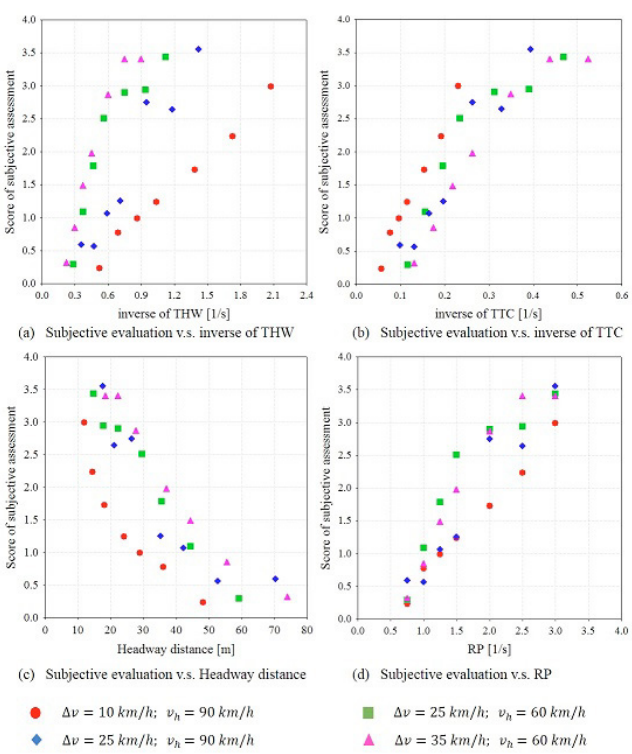

Fig. 8. Plots of subjective assessment versus THW_0, TTC_0, RP_0 and d_0

4. CONCLUSION

This study collected lane-changing data under different traffic conditions. Driver's lane-change *choice* and *implementation* have been investigated on measures of time-headway, time-to-collision, RP and headway-distance. Results show that:

- drivers tend to prefer time-to-collision information for perceiving driving risk for making a choice whether it is possible to change lanes in current traffic conditions with comparison to traffic flow like time-headway and headway-distance;
- drivers' lane-changing implementations are greatly dependent on subjective risk perception through time-to-collision and time-headway information.
- though drivers prefer time-to-collision and time-headway information for lane-change *choice* and *implementation* to RF, their subjective estimations of driving risk are consistent with the index of RF.

Since the index of RP involves two concepts of time-to-collision and time-headway information, it is believed that RP would be a significant index to quantify driver's subjective perception of proximity risk by a rear vehicle through optimizing weights on TTC and THW.

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